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Definition of the Hypothalamic GnRH Pulse Generator in mice

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Abstract

The pulsatile release of luteinizing hormone (LH) is critical for mammalian fertility. However, despite several decades of investigation, the identity of the neuronal network generating pulsatile reproductive hormone secretion remains unproven. We use here a variety of optogenetic approaches in freely-behaving mice to evaluate the role of the arcuate nucleus kisspeptin (ARN^{KISS}) neurons in LH pulse generation. Using GCaMP6 fibre photometry, we find that the ARN^{KISS} neuron population exhibits brief (~1 min) synchronized episodes of calcium activity occurring as frequent as every 9 min in gonadectomized mice. These ARN^{KISS} population events were found to be near-perfectly correlated with pulsatile LH secretion. The selective optogenetic activation of ARN^{KISS} neurons for 1 min generated pulses of LH in freely-behaving mice, whereas inhibition with archaerhodopsin for 30 min suppressed LH pulsatility. Experiments aimed at re-setting the activity of the ARN^{KISS} neuron population with halorhodopsin were found to re-set on-going LH pulsatility. These observations indicate the ARN^{KISS} neurons as the long-elusive hypothalamic pulse generator driving fertility.

Significance statement

Neural networks located in the hypothalamus are responsible for generating ultradian patterns of hormone secretion that control a wide variety of functions. How these neural networks generate pulsatile hormone secretion remain unknown. We report here that a population of hypothalamic kisspeptin neurons represent the gonadotropin-releasing hormone (GnRH) pulse generator. These cells have the remarkable ability to generate synchronized GnRH secretion every 9 minutes to drive pulsatile gonadotropin hormone secretion in the blood. These observations indicate the arcuate kisspeptin neurons as the origin of reproductive hormone pulsatility in mice and offer the prospect of better understanding and manipulating fertility in the clinic.

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Endocrine signaling from the pituitary gland occurs primarily as pulsatile hormone secretion. For most hormones, this pulsatility is thought to be generated by episode generators located within the hypothalamus and pituitary gland (1, 2). However, with few exceptions, it has been extremely difficult to determine the identity and nature of these neuroendocrine pulse generators (3, 4). Although the critical importance of pulsatile hormone secretion for reproduction was discovered in the 1970s (5, 6), the identity of the so-called “gonadotropin-releasing hormone (GnRH) pulse generator” (7, 8) responsible for pulsatile gonadotropin secretion remains unknown.

The discovery of the critical importance of kisspeptin signalling within the brain of humans and rodents for fertility (9-11) has suggested a potential role for kisspeptin signaling in the generation of pulsatile hormone secretion. Humans and rodents with global deletions or mutations in kisspeptin or the kisspeptin receptor fail to exhibit normal pulsatile luteinizing hormone (LH) secretion (9, 10, 12, 13). Of the two kisspeptin neuron populations that innervate GnRH neurons, particular attention has focussed upon the arcuate nucleus kisspeptin (ARN^{KISS}) neurons in terms of pulse generation (14, 15). Investigators have shown that ARN^{KISS} neurons co-express a variety of different neurotransmitters including glutamate, neurokinin B and dynorphin (15, 16), for which these cells also express functional receptors (17, 18). Alongside anatomical evidence, it has been suggested that the ARN^{KISS} neurons may exhibit synchronized behaviour that could intermittently activate GnRH neurons and, thus, act as a pulse generator (15, 16, 19). This attractive hypothesis remains unproven.

We employ here a range of optogenetic approaches in freely-behaving mice to evaluate whether the ARN^{KISS} neurons represent the GnRH pulse generator. Using GCaMP fiber photometry, we observed brief repetitive episodes of elevated calcium within the ARN^{KISS} neuron population and find a near-perfect correlation of these events with pulsatile LH secretion. Mimicking these brief synchronization events through selective channelrhodopsin (ChR2)-activation of ARN^{KISS} neurons was sufficient to generate pulsatile LH secretion while optogenetic inhibition of ARN^{KISS} neurons suppressed pulsatile LH secretion. Further, re-setting the activity of the ARN^{KISS} neuron population was found to re-set pulsatile LH secretion. Together these observations indicate that the ARN^{KISS} neurons are the GnRH pulse generator.

RESULTS

ARN^{KISS} neurons exhibit episodic activity in vivo

We used fiber photometry (20) to examine whether ARN^{KISS} neurons exhibited intermittent synchronous activity in awake freely behaving mice. We targeted the calcium reporter GCaMP6s to ARN^{KISS} neurons by bilateral injection of an AAV9-CAG-Flex-GCaMP6s into the ARN of male KISS1-Cre;tdTomato reporter mice (21). Immunohistochemical analyses showed that GCaMP6 was expressed in kisspeptin neurons throughout the ARN with 51-67% of ARN^{KISS} neurons expressing GCaMP6 depending on rostro-caudal level (Suppl Fig.1). Dual-labeled cells represented 89±6%